AQUAINTEL

AI-Powered Autonomous Prawn Farming and Disease Risk Management System

SOFTWARE IMPLEMENTATION AUDIT & TECHNICAL VERIFICATION REPORT (v2)

Audit Scope:	Rigorous second-pass implementation audit across Frontend, Backend, Database, AI/ML, IoT, and Financial Models.
Repository Location:	/Users/jeeshan/Documents/projects/aquaintel (Strictly isolated from peer directories)
Git Branch & Head:	main / commit 05a6f63 (Working tree verified clean)
Audit Date & Platform:	October 31, 2026 • macOS Darwin (arm64), Python 3.15.1, Node.js v24.22.1, SQLite 3
Software Readiness:	**READY FOR CONTROLLED SOFTWARE DEMONSTRATION** (32/32 tests passed, 0 build errors)
Physical Readiness:	**PHYSICAL DEPLOYMENT NOT VALIDATED — FIELD PILOT REQUIRED**
Supporting Evidence:	Machine-readable evidence files archived in reports/audit_evidence/

AUDITOR STATEMENT & SCOPE NOTICE: *This second-pass technical report reflects verified code artifacts and machine test results. It does not imply third-party certification or regulatory accreditation. Simulated subsystems are explicitly labeled as such, and financial models are presented as illustrative techno-economic projections rather than empirical farm receipts.*

Chapter 1 — Executive Summary & Audit Baseline

AquaIntel is a modular full-stack platform designed around the biological operations of commercial *Litopenaeus vannamei* farming in Nellore, Andhra Pradesh. This revised audit verifies that the codebase executes reliably as a software system while identifying critical prerequisites before physical farm deployment.

Subsystem Audit Status Matrix

Subsystem	Claimed Feature	Verified Evidence & Finding	Audit Classification
Backend API	17 REST Routers, JWT Auth, WebSockets	32/32 tests passed in 2.99s. All routes resolved under /api/v1.	**VERIFIED WORKING**
Frontend SPA	17 pages, Recharts telemetry, event injection	tsc -b & vite build passed (0 errors). oxlint passed (0 errors, 153 warnings).	**VERIFIED WORKING**
Database	10 ponds, 70 sensors, 40 aerators, seed data	Verified 10 ponds, 105 sensors, 45 aerators, 1,720 readings in aquaintel.db.	**VERIFIED WORKING**
AI Disease ML	Multi-pathogen Random Forest classifier	Trained on 1,200 synthetic rows; 55% acc; fails on EHP (0.0 F1) and EMS (0.056).	**SIMULATED (SYNTHETIC)**
Computer Vision	Check-tray feed and shrimp vigor detection	OpenCV contour detection with synthetic bounding box HUD overlay.	**PARTIALLY IMPLEMENTED**
IoT Edge Safety	Autonomous local hysteresis & feeder cutoff	State machine operates locally without internet in automated pytest.	**VERIFIED (SIMULATED)**
Physical Hardware	ESP32 buoys, LoRa SX1276 radios, RS485 probes	C++ firmware sketch present. Physical hardware not connected or tested.	**NOT VALIDATED (NO HW)**
Financial Returns	■22L CAPEX, ■1.56 Cr benefit, 1.7-mo payback	Mathematical model based on unverified disease & FCR assumptions.	**ILLUSTRATIVE MODEL**

Chapter 2 — Frontend & Backend Verification

Frontend Results: The React 19 application compiles cleanly (npm run build: 0 errors in 402ms). All 17 pages resolve via AppContext. Oxlint reports 0 errors and 153 minor non-blocking warnings (archived in reports/audit_evidence/frontend_audit.txt).
Backend Results: FastAPI service was verified through automated pytest execution (32 total / 32 passed / 0 failed in 2.99s). Dual health endpoints (/health and /api/health) return HEALTHY status. Test logs are archived in reports/audit_evidence/test_results.txt.

Automated Test Suite Evidence Breakdown

Pytest Module File	Scope Verified	Count	Result
test_ai_dataset_pipeline.py	ZIP discovery, CSV schemas, synthetic provenance, ML train/inference	10	**10 PASSED**
test_control_and_ai.py	Hypoxic emergency aeration, feeding cutoff, anomaly detector, offline state machine	6	**6 PASSED**
test_foundation.py	SQLite database connectivity, tables seed, health endpoints, farm overview	7	**7 PASSED**
test_phase2_core.py	API v1 routes, JWT login/me, telemetry ingestion, command lifecycle, pond CRUD	9	**9 PASSED**
TOTAL	**Automated Unit & Integration Coverage**	**32**	**32 PASSED (100%)**

Chapter 3 — Database Schema & Record Verification

Direct SQL examination of aquaintel.db verified 19 tables synced to Alembic migration revision 930fec5b847c. The database exceeds initial baseline claims: 10 ponds, 105 sensors (vs 70 baseline), 45 aerators (vs 40 baseline), 15 feeders, 14 users, and 1,720 sensor

readings. Full record inventory is archived in reports/audit_evidence/database_audit.json.

Chapter 4 — AI/ML & Disease Dataset Critical Review

DATASET PROVENANCE AUDIT: File synthetic_demo_data_NOT_FOR_VALIDATED_TRAINING.csv (1,200 rows × 28 cols) was examined. **100% of rows are synthetic** (data_source: SYNTHETIC, label_status: SYNTHETIC_LABEL). There are **0 real laboratory PCR confirmed records** (lab_test_type, lab_test_date, lab_test_result are 100% null). Full audit evidence is archived in reports/audit_evidence/dataset_audit.json.
ML MODEL EVALUATION: The Balanced Random Forest classifier (disease_risk_pipeline.joblib) was evaluated on a 20% holdout split (240 synthetic rows). Overall accuracy is **55.0%** (macro F1: **0.2314**). Model metrics are archived in reports/audit_evidence/ml_metrics.json.

Detailed Class-by-Class Metric Audit

Target Pathogen Class	Dataset Count	Precision	Recall	F1-Score	Audit Assessment
HEALTHY	613 (51.1%)	0.700	0.911	**0.792**	Moderate true-negative detection.
WSSV (White Spot)	177 (14.8%)	0.324	0.314	**0.319**	High false-positive rate (~68%).
VIBRIOSIS	122 (10.2%)	0.200	0.208	**0.204**	Low precision (80% false alarms).
BLACK_GILL	74 (6.2%)	0.333	0.200	**0.250**	Poor sensitivity (20% recall).
AHPND_EMS	121 (10.1%)	0.083	0.042	**0.056**	**Severely Deficient:** Only 1/24 caught.
MIXED_OR_UNRESOLVED	50 (4.2%)	0.000	0.000	**0.000**	**Complete Failure:** 0% detected.
EHP Microsporidian	43 (3.6%)	0.000	0.000	**0.000**	**Complete Failure:** 0% detected.
MACRO AVERAGE	**1,200 Rows**	**0.234**	**0.239**	**0.231**	**Unsuitable for autonomous clinical diagnosis.**

CLINICAL INTEGRITY NOTICE: *Synthetic classification accuracy does not demonstrate real-world disease prediction. Water quality parameters indicate physiological stress vulnerability, not pathogen presence. Veterinary pathology and RT-qPCR molecular laboratory tests are mandatory before making farm intervention decisions.*

Chapter 5 — IoT & Hardware Verification Breakdown

To maintain strict engineering rigor, this audit separates source code inspection and software simulation from physical microcontroller and field deployment testing:

Verification Tier	Target Component	Audit Method	Verified Finding & Status
Source Code Inspection	ESP32 Firmware (esp32_pond_master.ino)	Static code analysis	**VERIFIED:** 117 lines of C++ with RS485 Modbus, SX1276 LoRa, 10-min anti-cycle timer.
Automated Simulation	Edge Controller (edge_controller.py)	Pytest unit testing	**VERIFIED:** State machine cuts feeding and forces aeration during DO < 3.0 in simulation.
Gateway Simulation	Central LoRa Gateway (lora_gateway.py)	Python module inspection	**VERIFIED:** IN865 band packet framing and offline queue store-and-forward tested.
Physical Microcontrollers	ESP32-WROOM-32D & SX1276 Hardware	Bench hardware test	**NOT VALIDATED:** Physical microcontrollers were not connected during software audit.
Real Farm Deployment	Commercial 10-Ha Ponds (Nellore)	Field pilot trial	**NOT VALIDATED:** System has not been deployed on physical shrimp ponds.

Chapter 6 — Review of Financial & Techno-Economic Calculations

The repository documents a 10-hectare commercial model claiming an initial CAPEX of ■22 Lakhs, an annual net benefit of ■1.56 Crores, and a payback of 1.69 months (~1.7 months). This audit re-calculated the mathematical formulas and reviewed their underlying assumptions. Audit evidence is archived in reports/audit_evidence/financial_model.json.

Step-by-Step Mathematical Formula Verification

Financial Metric	Calculation Formula & Underlying Inputs	Modeled Result	Audit Classification
Initial CAPEX	10 DO/Temp probes (■4.5L) + 10 pH (■1.2L) + 10 Salinity (■1.8L) + 10 RTUs (■1.4L) + LoRa GW (■65K) + 10 Feeders (■5.2L) + 10 VFD Starters (■4.8L) + Server (■1.45L) + Installation (■1.0L)	■22,00,000 (■2.2L / Ha)	Illustrative Estimate (Equipment Quotes)
Annual OPEX	Sensor caps/reagents (■75K) + 4G SIMs (■24K) + Mechanical servicing (■85K) + Spares (■36K) = 10% of CAPEX	■2,20,000 / yr	Illustrative Estimate
Feed Savings	Production (187,500 kg) × FCR diff (1.50 - 1.25 = 0.25) × Feed price (■88/kg)	■41,25,000 / yr	Unverified Assumption (Requires field trial)
Energy Savings	Baseline (180,000 kWh) × 24% smart aeration reduction × ■7.50/kWh tariff	■3,24,000 / yr	Modeled Estimate (Diurnal DO curves)
Labor Savings	12 baseline laborers @ ■18K/mo (■25.92L/yr) × 30% automation reduction	■7,77,600 / yr	Illustrative Estimate
Disease Protection	Gross revenue (■7.875 Cr) × 13.5% assumed mortality reduction (18% down to 4.5%)	**■1,06,31,250 / yr** (67.9% of benefit)	**UNVERIFIED ASSUMPTION** (No clinical trial)
Net Annual Benefit	Gross benefit (■1,58,57,850) - Annual OPEX (■2,20,000)	**■1,56,37,850 / yr**	**Illustrative Projection**
Payback Period	CAPEX (■22,00,000) ÷ (Net Benefit ■1,56,37,850 ÷ 12 months)	**1.688 Months (~1.7 Months)**	**Modeled Projection**

CRITICAL FINANCIAL AUDIT VERDICT: *Notice that* ***67.9% of the projected financial return (■1.06 Crores out of ■1.56 Crores) hinges entirely on the assumption of a 13.5% reduction in catastrophic disease loss****. Because the disease ML model has not been validated on real clinical field data, this savings figure cannot be verified. These calculations represent an exploratory techno-economic model and must not be cited as guaranteed or verified earnings.*

Chapter 7 — Issues, Engineering Recommendations & Roadmap

Finding & Evidence	Severity	Engineering Impact	Required Action & Roadmap
Synthetic Disease Dataset: 100% of 1,200 rows synthetic; 0 real PCR lab tests.	**HIGH**	Classifier yields 55% acc; fails on EHP (0.0 F1) and AHPND (0.056 F1).	Partner with regional aquaculture research stations to collect RT-qPCR samples using real_farm_data_template.csv.
Unverified Financial Claims: ■1.56 Cr benefit assumes unvalidated disease protection.	**MEDIUM**	Risk of overpromising commercial returns to farm operators or investors.	Clearly classify financial models as exploratory projections; measure empirical FCR/energy deltas in field pilots.
Unrestricted CORS & Rate Limiting: allow_origins=['*'] and no rate limiter in FastAPI.	**MEDIUM**	Potential vulnerability to cross-site scripting and API request flooding.	Restrict CORS to production domain and configure slowapi rate limiting.
SQLAlchemy utcnow() Warning: 38 Pytest warnings regarding deprecated datetime calls.	**LOW**	Deprecation warnings in Python 3.12+; future breaking change.	Migrate model default timestamps to lambda: datetime.now(timezone.utc).

Final Audit Conclusion

CONCLUSION: AquaIntel demonstrates high technical competency in its software architecture, achieving clean TypeScript builds, passing 100% of its automated test suite (32/32), and enforcing defensive edge controller hysteresis loops.

• **Software Demonstration Readiness:** **READY FOR CONTROLLED SOFTWARE DEMONSTRATION** (Academic review, project evaluation, hackathons).

• **Physical Deployment Readiness:** **PHYSICAL DEPLOYMENT NOT VALIDATED**. Physical microcontroller testing, sensor bench calibrations, and clinical RT-qPCR data collection remain mandatory prerequisites.